

THOMAS HANNA

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Professional Summary

Detail-oriented Software Engineer with 4+ years of experience working with enterprise systems, APIs, and data-driven applications. Strong background in integrating front-end and back-end systems using RESTful web services, handling data workflows, and optimizing system performance. Experienced in collaborating with cross-functional stakeholders to build scalable solutions and troubleshoot production issues.

Professional Experience

New Jersey Institute of Technology, Jersey City, NJ

Jan 2025 - August 2026

Master of Science in Computer Science

Graduate Research & Engineering Focus (Data Systems, Machine Learning, Distributed Systems)

- Designed and implemented data-intensive applications and ML pipelines using Python, applying techniques in classification, clustering, and time-series analysis across academic datasets.
- Built backend services using FastAPI, exposing application functionality through RESTful APIs and enabling clear separation of frontend and backend systems.
- Explored scalable data mining algorithms (FP-tree, decision trees, association rule mining) and their tradeoffs in large-scale environments.
- Applied distributed systems and cloud concepts to design resilient and scalable architectures.
- Evaluated ML tools and cyberinfrastructure platforms for reproducibility, usability, and scalability, documenting challenges and optimization strategies.
- Prepared applications for containerization and cloud deployment using Docker and modern DevOps practices

Tech Stack: Python, FastAPI, REST APIs, NumPy, SQL, Docker, AWS (S3, EC2), GitHub Actions

GENERAL ELECTRIC CORPORATE, Atlanta, GA (Remote)

May 2022 - November 2024

Software Engineer

Led modernization and performance optimization of GE's global order management platform.

- Designed and integrated front-end applications with backend APIs and enterprise systems, supporting 10K+ internal users across global business units.
- Developed and maintained RESTful API integrations, ensuring reliable data exchange between services and improving system responsiveness by 30%.
- Collaborated with cross-functional stakeholders (product, UX, backend teams) to define system requirements and integration touchpoints.
- Troubleshoot production issues related to API failures, data inconsistencies, and system performance, reducing incident resolution time by 25%.
- Built reusable components and services that standardized data handling and improved maintainability across teams.
- Supported CI/CD pipelines and cloud deployments, ensuring stable and consistent system integrations.

Tech Stack: APIs, REST, TypeScript, React, AWS, Docker, SQL, Azure

INNCRETECH, Princeton, NJ

August 2020 - May 2022

Software Engineer

Developed and maintained scalable, responsive web platforms serving clients across logistics, retail, and healthcare sectors.

- Integrated front-end applications with backend services (Java/Spring Boot), improving data retrieval performance by 25%.
- Worked with APIs and structured data sources to support business workflows across logistics, retail, and healthcare platforms.
- Collaborated with stakeholders to translate business requirements into technical solutions and system integrations.
- Assisted in debugging and resolving system issues related to data flow, API responses, and application performance.
- Participated in testing and validation of data accuracy and system reliability across environments.

Tech Stack: Java, Spring Boot, SQL, REST APIs, Node.js, Postman

Education

Bachelor of Science, BS, Computer Science

Rutgers University, New Brunswick, NJ

Masters of Science, MS, Computer Science

NJIT, Newark, NJ

Core Strengths

React & TypeScript Architecture • Frontend Performance Optimization • Component Library Development

Next.js / SPA Design • Migration (Angular → React) • ADA Compliance & SEO • Cross-Team Collaboration

Testing Automation (Jest, Cypress) • CI/CD with Docker, AWS, and Azure • Mentorship & Technical Hiring

Technical Skills

Languages: JavaScript, TypeScript, Java, Python, SQL

Integration & APIs: REST APIs, Web Services, JSON, Postman

Frameworks: React, Node.js, Spring Boot

Databases: MySQL, PostgreSQL, MongoDB

Tools: Git, Docker, CI/CD, AWS, Azure

Project Highlights

Enterprise Migration Project: Migrated 50K+ lines of AngularJS code to React and TypeScript, eliminating technical debt and cutting feature delivery time by 30%.

Component Library Initiative: Built 60+ React UI components adopted across 3 internal teams, standardizing UI and reducing duplicated code by 28%.

Performance Optimization: Reduced page load times from 4.2s to 2.1s via code splitting, caching, and image compression.

Accessibility Overhaul: Achieved 100% ADA/WCAG compliance across multiple applications using semantic HTML and ARIA attributes.